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Hallucination Detection in Large Language Model Generated Texts: A Survey of Internal and External Approaches

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***Abstract*—For large language models (LLMs) to guarantee the trustworthiness and consistency of data, hallucination detection is essential. According to current research, the LLM model can generate false and inaccurate data that compromises the models' dependability. A review on hallucination detection in LLMs was given in this paper. This study distinguished between internal and external methods for detecting hallucinations. While external models rely on external tools, internal approaches rely on signals within the model. In addition to outlining future research possibilities, this work describes methodological categories, frameworks, applications, and distinctions between various hallucination detection approaches. This article offers organized knowledge about reliable LLM model developments and hallucination detection.**

***Index Terms*—LLM, NLP, Reliability, Trustworthiness, Hallucination Detection, RAG, AI.**

I. Introduction

Large Language Models (LLMs) are currently tested in different text, image, and data processing applications, such as text summarization, sentiment analysis, virtual assistants, and content creation. Healthcare, banking, law, and education are just a few of the fields that use LLM models extensively [1], [2], [3], [4]. However, there were serious problems with LLM models, such as factual reliability or hallucinations, large computational costs, and problems with reasoning and ethics. When an LLM experiences hallucinations, it generates material that looks authoritative and well-written, but it contains false information, contradicting assertions, factual errors, or unreliable information. Current surveys don't make a basic distinction between detection strategies that operate inside the LLM's architecture and those that need outside documents and knowledge sources to be verified [5], [6], [7], [8]. Internal model-based and external knowledge-based techniques are the two primary categories into which this research has separated hallucination detection methodologies in order to address these problems. This study examined methods that examine model confidence scores, attention patterns, hidden state representations, uncertainty measurements, and consistency over several generations for internal model-based approaches. This article investigated fact-checking against knowledge bases, retrieval-augmented verification, cross-referencing with reliable sources, and human-in-the-loop validation systems on the external knowledge-based side. Below is a list of this survey paper's noteworthy contributions:

- In order to automatically recognize hallucinations in writings generated by LLM, this work presents a single taxonomy that blends model-based and external knowledge-based approaches.
- This survey study offers a thorough overview of automated detection techniques in both categories, combining various approaches like fact-checking, retrieval-augmented verification, model confidence analysis, and hidden state representations.
- This work highlights important issues and potential avenues for future research while analyzing current standards and datasets.

Related research on hallucination detection for LLMs is summarized in Section II. The suggested classification scheme is shown in Section III. Within this paradigm, methodological categories are identified in Section IV. The main distinctions between current methods are hinted to in Section V. Future works and limits are offered in next Section. The key summary of this paper is included in last Section.

II. Related Works

Literary works pertaining to the identification of hallucinations in LLMs were discussed in this section. Using GPT and LLaMA models on English biographical and medical datasets, the authors in [1] proposed entropy-based uncertainty estimators and presented semantic entropy-based detection. With 0.89 AUROC, GPT-4 with semantic entropy clustering performed the best. Using the internal states of LLMs, the paper in [2] constructed unsupervised hallucination detection, which tested on GPT models, LLaMA-2, and Vicuna models with LLaMA-2-7B reaching near 87% accuracy. Using metamorphic relations and prompt mutations, the authors in [3] present a self-contained method for detecting hallucinations

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that works and achieves near 92% accuracy across GPT, Claude, and Gemini models without the need for external resources. In order to test ChatGPT, GPT-4, Claude, and Vicuna models, the work in [4] created HaluEval, a large-scale English benchmark covering a variety of tasks like question answering, knowledge-based dialogue, and text summarization. GPT4 showed superior performance with an average hallucination detection rate of near 82%. Larger models like GPT-3 (175B parameters) only achieved 58% accuracy compared to smaller T5-small (68% accuracy), according to the work in [5], which developed TruthfulQA, an adversarial benchmark for measuring truthfulness in English language models. In [6], the authors benchmarked for intrinsic hallucinations, investigated how models deviated from the input context, and addressed knowledge conflicts in the face of conflicting data.

III. Classification Framework for Hallucination Detection Approaches

The taxonomy that separates the main methods for detecting hallucinations in LLMs is shown in this section. These methods are classified into two main types: internal and external knowledge-based methods. This study listed representative techniques for each category and cited pertinent works that demonstrate them.

A. Internal Model-Based Detection

- **Uncertainty-Based Methods:** These uncertainty methods identify erroneous or hallucinogenic replies using internal confidence scores, probability, or entropy-based factors. For instance, semantic-level uncertainty metrics are used in both [1] and [14] to identify possible hallucinations. A uncertainty-based detector that utilizes both token features and reliability of keyword is discussed in [2].
- **Consistency-Based Approaches:** These type of models assess whether the model consistently produces responses to cues that are equal or semantically paraphrased. Stability across various generations is tested in [35] using sampler-based consistency techniques.
- **Attention Pattern Analysis:** These methods look for unusual patterns suggestive of hallucinations by analyzing hidden-state dynamics and attention heads. To find inconsistency signals, the work in [11] suggested a spectrum attention method.
- **Self-Reflection Techniques:** These techniques encourage LLMs to evaluate their own results and spot problems. A zero-shot detection technique that divides input prompts into attentive and non-attentive subsets and calculates consistency scores across answers was presented in [36].

B. External Knowledge-Based Detection

- **Knowledge Base Verification:** These methods use structured knowledge sources like Wikidata or knowledge bases (field wise) to cross-reference assertions produced by LLMs [28].
- **Retrieval-Augmented Verification:** These approaches evaluate whether LLM outcomes are grounded or not. For this, they retrieves supporting documents from external sources using information retrieval techniques [21].
- **Fact-Checking Integration:** These systems use databases or fact-checking tools to validate LLM claims. By contrasting generated assertions with reference text, the work in [20] identified factual errors in summaries.

IV. Methodological Categories

The section provides an organized summary of the methods for locating and reducing factually inaccurate outputs in LLMs. Comprehensive comparisons, identification methods, performance indicators, and dataset types are shown in Tables I and II.

A. Internal Model-Based

The internal based model suffers from some challenges like uncertainty quantification, interpretation, and model calibration. Embedding-based evaluation is one of the fundamental method in which models evaluate the consistency and quality of content using their own trained representations. The work in [7] assesses text production using BERT embeddings to capture semantic similarity beyond surface-level lexical overlap. The study in [8] created an unsupervised system for hallucination detection. They did not use any manual annotations.

The paper in [9] introduced logit-level detection to more accurately identify knowledge hallucinations while maintaining detection power through LLMs' internal states. Sriramanan and associates. Without further training in GPT or LLaMA, the study [10] suggested LLM-Check, which analyzes hallucination detection inside single LLM replies utilizing internal attention kernel maps. Recent research by [11] showed that using top-k eigenvalues with spectral properties and statistical analysis. They used attention maps to identify content that is real and hallucinogenic. Using Llama and Phi-3.5, it demonstrated the rich semantic representations seen in self-attention patterns across several tokens. The analysis in [12] demonstrated that using information from all layers rather than individual layers in isolation leads to better performance in hallucination detection tasks.

The work in [13] indicated that neurons along with tokens and activation method can affect the hallucination risk of LLMs. The work in [14] addressed the five-to ten-fold increase in computational costs associated with full semantic entropy computation by developing semantic entropy probes as reliable and computationally efficient. Through empirical examination of hidden states and the experimental methodology, the study [15] examined whether LLMs are aware of hallucinations. Through probing estimators, they achieve an average accuracy (hallucination) of near 84 percent at runtime. By conditioning just on input hidden states, the work in [16] are able to identify hallucinations and direct models toward more factual outputs when hallucinations are anticipated. This method is a major

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TABLE I
INTERNAL MODEL-BASED HALLUCINATION DETECTION METHODS

Method	Type	Architecture	Dataset	Evaluation Metrics	Reference
(i) Embedding-Based Systems					
BERTScore	Sequence-wise	BERT variants	Summarization corpora	Correlation with human judgments vs ROUGE	[7]
Sentence Embeddings	Sequence-wise	Transformer encoders	Multi-domain text	Cosine similarity scores	[12]
Multi-layer Features	Token + Sequence	Deep transformers	Various NLG tasks	Layer-wise performance analysis	[9]
Contrastive Learning	Sequence-wise	Encoder-decoder models	Factual/hallucinated pairs	Embedding discrimination metrics	[15]
(ii) Uncertainty Estimation Systems					
Semantic Entropy	Sequence-wise	GPT-4, LLaMA-2	Biographical generation	AUROC: 0.89 vs 0.73 traditional	[1]
Semantic Entropy Probes	Token-wise	Multiple LLMs	QA datasets	5-10x computation cost reduction	[14]
Token Probability	Token-wise	GPT-3, GPT-4	TruthfulQA, general QA	Confidence calibration curves	[17]
BSDetector	Sequence-wise	Various architectures	Multi-domain datasets	Precision, Recall, F1-score	[15]
Calibration Methods	Token + Sequence	Transformer models	Factual QA benchmarks	ECE, MCE, reliability diagrams	[18]
Entropy Detection	Token-wise	Multiple LLM families	Zero-shot evaluation	AUROC, detection accuracy	[13]
(iii) Internal State Analysis Systems					
INSIDE	Token-wise	Various LLMs	Knowledge-based QA	Logit-level accuracy	[9]
LLM-Check	Token + Sequence	Single LLM responses	Diverse hallucination settings	Baseline improvement metrics	[10]
Hidden State Probing	Sequence-wise	GPT, BERT variants	Labeled hidden states	84.32% average accuracy	[15]
Preemptive Detection	Sequence-wise	Multiple architectures	Pre-generation analysis	Early intervention success rate	[16]
Temporal Dynamics	Token-wise	Autoregressive models	Time-series generation	Pattern evolution tracking	[8]
(iv) Attention Analysis Systems					
Spectral Attention	Token-wise	Transformer models	Truthful vs hallucinated pairs	Spectral feature discrimination	[11]
Multi-head Analysis	Token + Sequence	BERT, GPT families	Multi-task evaluation	Cross-head consistency metrics	[12]
Attention Kernels	Token-wise	Various LLMs	Single response analysis	Kernel pattern recognition	[10]
Layer-wise Attention	Sequence-wise	Deep transformers	Layered attention datasets	All-layer vs individual performance	[11]
(v) Self-Consistency Systems					
Multiple Sampling	Sequence-wise	GPT-3, GPT-4, Claude	Consistency benchmarks	Variability-based metrics	[2]
Self-Assessment	Sequence-wise	Various LLMs	Self-evaluation datasets	Meta-reasoning accuracy	[15]
Response Variation	Sequence-wise	Multiple architectures	Statistical consistency data	Variance-based detection	[17]
(vi) Hybrid Methods					
Multi-signal Fusion	Token + Sequence	State-of-the-art LLMs	Comprehensive benchmarks	Combined metric optimization	[8]
Ensemble Detection	Sequence-wise	Various model families	Cross-domain evaluation	Robust performance metrics	[10]

step in preventing hallucinations rather than just identifying them. Ensemble techniques and multi-layer feature extraction have demonstrated potential for enhancing detection robustness. Sequence based techniques in [18] give more reliable detection by taking into account wider contextual patterns and token-level approaches in [17] offer more precise identification features.

For the purpose of detecting hallucinations, the work in [19] ICR employs a representation-based and hidden state approach. Their solution captures sequence-level crosslayer dynamics to better differentiate true from hallucinated content, in contrast to other static approaches. Using common factuality standards like TruthfulQA, QA datasets, and summarization tasks, the authors assessed the method on GPT-3 and GPT-4 models.

B. External Knowledge-Based

Weakly-supervised learning is a fundamental method in external knowledge-based detection for detecting factual consistency. Kryscinski and associates. FactCC, which uses weakly-supervised BERT and RoBERTa classifiers trained on artificial positive and negative instances of abstractive text summarization, was created by [20]. Their RoBERTa-large model outperformed conventional lexical metrics like F1 score of near 71 percent. Using BART-large for question generation and BERT-large for answer extraction, the inspection [21] introduced QAGS, which creates questions from summaries and compares responses when the questions are posed to both summaries and source articles. Furthermore, [22] introduces non-fabrication and multi-step verification. A complex method for external knowledge-based detection and mitigation is RAG systems. The study [23] suggested SAFE approach, an assessment technique for verifying long-form factuality in LLMs that greatly lessens hallucinations by using retrieval to base model development. In these systems, the retriever module is in charge of extracting pertinent sections or documents from sizable corpora in response to input queries. The information that is recovered is then sent to the LLM, which uses prompts and external knowledge to generate final responses [25]. A further study [26] presented a RAG framework for math question-answering in which prompts for an unidentified OpenAI LLM are injected with texts taken from an OpenStax Propregebra

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TABLE II
EXTERNAL KNOWLEDGE-BASED HALLUCINATION DETECTION METHODS

Method	Type	Architecture	Dataset	Evaluation Metrics	Reference
(i) Factual Consistency Detection Systems					
FactCC	Sequence-wise	BERT, RoBERTa classifiers	CNN/DailyMail, XSum	F1 score: 71.6% on human annotations	[20]
QAGS	Sequence-wise	BART-large, BERT-large	CNN/DailyMail, XSum	Pearson correlation: 0.71 with human judgments	[21]
FEQA	Sequence-wise	T5, BERT variants	Summarization datasets	Question-answer agreement scores	[30]
SummaC	Sequence-wise	RoBERTa, ELECTRA	CNN/DailyMail, XSum, QAGS-XSum	Balanced accuracy, AUC scores	[32]
(ii) Knowledge-Grounded Verification Systems					
KnowHalu	Sequence-wise	GPT-4, LLaMA variants	Wikipedia, multi-domain QA	15.65% improvement in QA, 5.50% in summarization	[22]
SAFE	Document-wise	Retrieval + LLM evaluator	Long-form generation datasets	Factuality evaluation scores	[24]
FacTool	Multi-modal	Tool-augmented framework	Multi-task, multi-domain	Task-specific accuracy metrics	[32]
Self-Checker	Sequence-wise	Plug-and-play modules	Fact-checking datasets	Precision, recall, F1 for fact verification	[33]
(iii) Retrieval-Augmented Detection Systems					
RAG Detection	Document-wise	Dense retriever + BART/T5	OpenQA, Wikipedia corpus	Retrieval accuracy, generation faithfulness	[25]
MultiRAG	Multi-source	Multiple retrievers + LLM	Cross-domain knowledge bases	Multi-source agreement metrics	[27]
RetroMAE	Sequence-wise	Masked autoencoder retrieval	Web-scale corpora	Retrieval precision, hallucination reduction	[34]
REALM	Token-wise	BERT + dense retrieval	Open-domain QA datasets	Exact match accuracy, retrieval success	[25]
(iv) Benchmark and Evaluation Systems					
TruthfulQA	Sequence-wise	Multiple LLMs evaluation	817 adversarial questions, 38 categories	Truthfulness accuracy: GPT-3 58%, T5-small 68%	[5]
HaluEval	Multi-task	GPT-4, GPT-3.5, Claude	QA, dialogue, summarization	Detection rate: GPT-4 82.4% average	[4]
HaluEval 2.0	Multi-domain	Various LLMs	8,770 questions across 5 domains	Domain-specific hallucination rates	[29]
FEVER	Claim-wise	Evidence retrieval + classification	Wikipedia-based claims	FEVER score, evidence retrieval accuracy	[28]
HalluLens	Multi-granular	LLM evaluation framework	Intrinsic hallucination benchmarks	Knowledge conflict handling metrics	[6]
(v) Domain-Specific Detection Systems					
Medical Fact Check	Claim-wise	Domain-adapted BERT	Medical literature, PubMed	Clinical accuracy, evidence quality scores	[34]
Legal Verification	Document-wise	Legal knowledge base + LLM	Legal case databases	Citation accuracy, precedent verification	[34]
Scientific Claims	Citation-wise	SciBERT + knowledge graphs	Scientific publications	Citation accuracy, claim support metrics	[31]
(vi) Hybrid External-Internal Systems					
External + Confidence	Sequence-wise	Retrieval + uncertainty estimation	Multi-domain datasets	Combined external verification + confidence scores	[23]
Knowledge-Uncertainty	Token + Sequence	Knowledge base + internal states	Factual QA benchmarks	Hybrid detection accuracy metrics	[31]
Multi-signal External	Document-wise	Multiple knowledge sources + LLM	Comprehensive evaluation sets	Ensemble external verification scores	[27]

textbook. Human judges in this work favor RAG-enabled responses. However, these responses lose attractiveness when the prompting places an excessive emphasis on grounded textbook information. In order to answer factoid questions, the analysis in [27] integrates retrieved images and text from a large corpus. In order to provide baseline for fact extraction and verification, the work in [28] developed the FEVER dataset, which requires statements to be categorized as supported, rejected, or lacking adequate evidence based on Wikipedia articles. Important methodological guidelines for evidence-based fact checking were created by this dataset, which had an impact on later methods for detecting hallucinations. The study in [30] presents a fundamental investigation of hallucinations in English summarization systems, examining several models on CNN and XSum datasets, including BERT-based extractive systems, BART-large, and T5-base. Their research developed the intrinsic-extrinsic hallucination concept that is still in use today, with BART-large striking the ideal balance between 23 percent extrinsic hallucination rate and 78 percent fidelity. The work in [31] provides an overview of hallucinations in NLG by differentiating between intrinsic and extrinsic hallucinations. The advent of post-hoc fact-checking and verification systems as workable solutions for real-world deployment is covered in [32]. Current methods include LLM-Augmenter frameworks that enhance LLM output using external knowledge sources and truth-o-meter systems that use web mining to fact-check LLM output. Self-Checker framework for fact-checking employing LLMs like GPT-3 and ChatGPT, was introduced in

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[33].

V. Key Differences Among Methods

This study examined a large body of research on LLM methods for texts based hallucination identification. This study divided the current systems into two major groups: both external knowledge-based detection systems and internal model-based detection systems. This paper demonstrated the differences between them in terms of architectures, datasets, procedures, evaluation systems, and detection techniques. Internal model-based systems are computationally efficient for real-time deployment because they make use of the LLM's internal operations without requiring external knowledge sources. The majority of internal systems rely on hidden state probing, attention pattern analysis, self-consistency analysis, and uncertainty estimates. The aforemntioned systems used transformer based architectures (GPT models, LLaMA) and different kind of probing classifiers. Different granularities can be provided by both token-level and sequence-level analysis. It can noted that hybrid approaches can combine several internal signals and also yield the best results. Retrieval systems and KBs must be intricately integrated in external knowledge-based systems to compare created content with reliable sources. The main types of this systems are benchmark evaluation platforms, fact-checking platforms, and RAG platforms. Retrieval systems and KBs are often incorporated with some architectures such as BERT, T5, and GPT versions.

The evaluation approaches of internal and external systems differ significantly. AUROC, accuracy, precision-recall measures, and correlation coefficients with human assessments are commonly used to assess internal model-based systems. Quantitative evaluation skills are demonstrated by performance indicators such as near 84 percent accuracy for hidden state probing and 0.89 AUROC for semantic entropy. External systems use a wider range of evaluation techniques, such as task-specific accuracy measures, factuality evaluations, and F1 scores. Benchmark analyses show significant differences in performance, with GPT-4 attaining near 82 percent detection rates and smaller models occasionally outperforming bigger ones in truthfulness tests.

VI. Limitations and Future Work

Hallucination detection still suffers from significant obstacles despite moderate advancements. Most methods are unable to generalize beyond certain domains. It can be noted that external verification techniques are still computationally costly. The current benchmarks are dispersed and lack consistent evaluation. Biases are introduced by relying on outside resources. Internal uncertainty signals are frequently difficult for wider consumers to understand. Future research should focus on user-friendly interpretability, hybrid techniques incorporating internal and external clues. Further scopes for future research can include lightweight and accurate models, comprehensive benchmarks, spanning domains, and languages, and expanding detection to multimodal situations.

VII. Conclusion

The reliable implementation of LLMs in safety-critical and knowledge-intensive applications is still fundamentally hampered by hallucination detection. In addition to highlighting representative techniques under each area, this survey article offered an organized taxonomy of detection methods that distinguished between internal and external approaches. Although current approaches show encouraging outcomes, there are still large gaps in scalability, trustworthiness, generalization, ethics checking, validation, and evaluation. Interdisciplinary initiatives combining developments in NLP, knowledge representation, interpretability, and human-AI interaction are necessary to address these issues. Finally, developing reliable, accountable, and socially conscious AI systems requires efficient hallucination detection in addition to being a technical requirement.

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